

USER INFORMATION
ONDATRON
ONDANSETRON INJECTION BP 2mg/mL
Sterile and Non-Pyrogenic

Composition:

Each mL solution contains 2mg Ondansetron BP corresponding to 2.5mg Ondansetron Hydrochloride Dihydrate.
Each 2 mL ONDATRON contains 4 mg Ondansetron BP.
Each 4 mL ONDATRON contains 8 mg Ondansetron BP.

Characteristics:

Ondansetron Injection is a sterile and non-pyrogenic solution of Ondansetron Hydrochloride Dihydrate in Water for Injections. The solution is a clear and colourless solution.

Pharmacodynamics:

Ondansetron is a potent, highly selective 5HT₃ receptor-antagonist. Its precise mode of action in the control of nausea and vomiting is not known. Chemotherapeutic agents and radiotherapy may cause release of 5HT in the small intestine initiating a vomiting reflex by activating vagal afferents via 5HT₃ receptors. Ondansetron blocks the initiation of this reflex. Activation of vagal afferents may also cause a release of 5HT in the area postrema, located on the floor of the fourth ventricle, and this may also promote emesis through a central mechanism. Thus, the effect of ondansetron in the management of the nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy is probably due to antagonism of 5HT₃ receptors on neurons located both in the peripheral and central nervous system. The mechanisms of action in post-operative nausea and vomiting are not known but there may be common pathways with cytotoxic induced nausea and vomiting. Ondansetron does not alter plasma prolactin concentrations.

Pharmacokinetics:

Peak plasma concentrations of ondansetron occur about 1.5 hours after an oral dose of 8 mg, and about 6 hours after a rectal dose. The absolute bioavailability is about 60%, mainly because of hepatic first-pass metabolism. In elderly subjects, bioavailability may be somewhat higher (65%) and clearance lower, presumably due to reduced hepatic first-pass metabolism. Ondansetron is extensively distributed in the body; about 70 to 75% of the drug in plasma is protein bound. It is metabolised in the liver through multiple enzymatic pathways; ondansetron is a substrate for cytochrome P450 isoenzymes, primarily CYP3A4, but also CYP1A2 and CYP2D6. Less than 5% of a dose is excreted unchanged in the urine. The terminal elimination half-life is about 3 hours after oral or parenteral doses, and about 6 hours after rectal use. The terminal elimination half-life is prolonged to about 5 hours in the elderly and in those with renal impairment. These differences are not considered sufficient to warrant dosage adjustment. However, in patients with severe hepatic impairment, bioavailability may approach 100% and clearance is markedly reduced, with elimination half-lives of 15 to 32 hours; dosage restriction is advisable. In general, children have a higher clearance compared with adults, although age-related reductions in clearance have also been reported, with younger children having lower clearances. Use of weight-based doses compensates for these changes and normalises exposure in paediatric patients.

Indication:

Adults:
Ondansetron injection is indicated for the management of nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy. Ondansetron injection is also indicated for the prevention and treatment of post-operative nausea and vomiting.
Paediatric Population Injection and oral formulations:
Ondansetron is indicated for the management of nausea and vomiting induced by cytotoxic chemotherapy. No studies have been conducted on the use of orally administered ondansetron in the prevention or treatment of post-operative nausea and vomiting; IV injection is recommended for this purpose.

Recommended Dosage:

CHEMOTHERAPY AND RADIOTHERAPY INDUCED NAUSEA AND VOMITING (CINV AND RINV)

The emetogenic potential of cancer treatment varies according to the doses and combinations of chemotherapy and radiotherapy regimens used. The selection of dose regimen should be determined by the severity of the emetogenic challenge.

Populations

CINV and RINV in Adults

The recommended intravenous (IV) or intramuscular (IM) dose of ondansetron is 8 mg administered immediately before treatment. A single dose of intravenous ondansetron given for the prevention of CINV in adults (aged less than 75 years) must not exceed 16mg (infused over at least 15 minutes). The efficacy of ondansetron in highly emetogenic chemotherapy may be enhanced by the addition of a single IV dose of dexamethasone sodium phosphate 20 mg, administered prior to chemotherapy. IV doses greater than 8 mg and up to a maximum of 16 mg must be diluted in 50 mL to 100 mL of 0.9% Sodium Chloride Injection or 5% Dextrose Injection before administration and infused over not less than 15 minutes. Ondansetron doses of 8 mg or less do not need to be diluted and may be administered as a slow IM or IV injection in not less than 30 seconds. The initial dose of ondansetron may be followed by 2 additional IV or IM doses of 8 mg 2 to 4 hours apart, or by a constant infusion of 1 mg/h for up to 24 hours. Oral treatment is recommended to protect against delayed or prolonged emesis after the first 24 hours.

CINV in Children and Adolescents (aged 3 years to 17 years)

The dose for CINV can be calculated based on body surface area (BSA) or weight. In paediatric clinical studies, ondansetron was given by IV infusion diluted in 25 to 50 mL of saline or other compatible infusion fluid and infused over not less than 15 minutes.

Dosing by BSA

Ondansetron should be administered immediately before chemotherapy as a single IV dose of 5 mg/m². The IV dose must not exceed 8 mg. Oral dosing can commence 12 hours later and may be continued for up to 5 days (Table 1). Adult doses must not be exceeded.

Table 1. BSA-based dosing for CINV

BSA	Day 1	Days 2 - 6
≥0.6 m ² to ≤1.2 m ²	5 mg/m ² IV plus 4 mg tablet after 12 hours	4 mg tablet every 12 hours
>1.2 m ²	5 mg/m ² IV or 8 mg IV plus 8 mg tablet after 12 hours	8 mg tablet every 12 hours

Dosing by body weight

Ondansetron should be administered immediately before chemotherapy as a single IV dose of 0.15 mg/kg. The IV dose must not exceed 8 mg. On Day 1, two further IV doses may be given in 4-hourly intervals. Oral dosing can commence 12 hours later and may be continued for up to 5 days (Table 2). Adult doses must not be exceeded.

Table 2. Weight-based dosing for CINV

Body Weight	Day 1	Days 2 - 6
> 10 kg	Up to 3 doses of 0.15mg/kg IV every 4 hours	4 mg tablet every 12 hours

There are no dosing recommendations for children with BSA <0.6 m², body weight <10 kg or who are unable to swallow tablets.

CINV and RINV in Elderly

Ondansetron is well tolerated by patients over 65 years of age. In patients 65 years of age or older, all IV doses should be diluted and infused over 15 minutes and, if repeated, given no less than 4 hours apart. In patients 65 to 74 years of age, the initial IV dose of ondansetron 8 mg or 16 mg, infused over 15 minutes, may be followed by 2 doses of 8 mg infused over 15 minutes and given no less than 4 hours apart. In patients 75 years of age or older, the initial IV dose of ondansetron should not exceed 8 mg infused over 15 minutes. The initial dose of 8 mg may be followed by 2 doses of 8 mg, infused over 15 minutes and given no less than 4 hours apart. All intravenous doses of ondansetron for the prevention of CINV should be diluted in 50–100mL saline or other compatible fluid and infused over at least 15 minutes.

Repeat dosing in all adult patients (including elderly patients)

Repeat intravenous doses of ondansetron for the prevention of CINV should be given no less than 4 hours apart.

Post-Operative Nausea and Vomiting (PONV)

PONV in Adults

For prevention of post-operative nausea and vomiting, the recommended dose of ondansetron injection is a single dose of 4 mg by IM or slow IV injection administered at the induction of anaesthesia.

For treatment of established post-operative nausea and vomiting a single dose of 4 mg given by IM or slow IV injection is recommended.

PONV in Children and Adolescents (aged 3 years to 17 years)

For prevention and treatment of PONV in paediatric patients having surgery performed under general anaesthesia, ondansetron may be administered by slow IV injection (not less than 30 seconds) at a dose of 0.1 mg/kg up to a maximum of 4 mg either prior to, at or after induction of anaesthesia, or after surgery.

Elderly

There is limited experience in the use of ondansetron in the prevention and treatment of post-operative nausea and vomiting in the elderly, however ondansetron is well tolerated in patients over 65 years receiving chemotherapy.

Renal Impairment in CINV, RINV & PONV

No alteration of daily dosage or frequency of dosing, or route of administration are required.

Hepatic Impairment in CINV, RINV & PONV

Clearance of ondansetron is significantly reduced and serum half-life significantly prolonged in subjects with moderate or severe impairment of hepatic function. In such patients a total daily dose of 8 mg IV or oral should not be exceeded.

Patients with Poor Sparteine/Debrisoquine Metabolism in CINV, RINV & PONV

The elimination half-life of ondansetron is not altered in subjects classified as poor metabolisers of sparteine and debrisoquine. Consequently in such patients repeat dosing will give drug exposure levels no different from those of the general population. No alteration of daily dosage or frequency of dosing is required.

Route of Administration: Intravenous(IV)/Intramuscular(IM)

Contraindications:

Based on reports of profound hypotension and loss of consciousness when ondansetron was administered with apomorphine hydrochloride, concomitant use with apomorphine is contraindicated. Hypersensitivity to any component of the preparation.

Warnings and Precautions:

Hypersensitivity reactions have been reported in patients who have exhibited hypersensitivity to other selective 5HT₃ receptor antagonists. Very rarely and predominantly with intravenous ondansetron, transient ECG changes including QT interval prolongation have been reported. As ONDATRON is known to increase large bowel transit time, patients with signs of subacute intestinal obstruction should be monitored following administration. Hypokalemia and hypomagnesemia should be corrected prior to ondansetron administration. Serotonin syndrome has been described following the concomitant use of Ondansetron and other serotonergic drugs. If concomitant treatment with Ondansetron and other serotonergic drugs is clinically warranted, appropriate observation of the patient is advised.

Caution:

Single dose container. Discard all unused contents. Do not use if the solution is discoloured and leakage is detected. If any visible solid particles, growth or turbidity appears during storage, the product should not be used.

Interactions with Other Medicaments:

There is no evidence that ONDATRON either induces or inhibits the metabolism of other drugs commonly coadministered with it. Specific studies have shown that there are no pharmacokinetic interactions when ONDATRON is administered with alcohol, temazepan, furosemide, tramadol or propofol.

Ondansetron is metabolised by multiple hepatic cytochrome P-450 enzymes:

CYP3A4, CYP2D6 and CYP1A2. Due to the multiplicity of metabolic enzymes capable of metabolising ondansetron, enzyme inhibition or reduced activity of one enzyme (e.g. CYP2D6 genetic deficiency) is normally compensated by other enzymes and should result in little or no significant change in overall ondansetron clearance or dose requirement.

Phenytoin, Carbamazepine and Rifampicin:

In patients treated with potent inducers of CYP3A4 (i.e. phenytoin, carbamazepine, and rifampicin), the oral clearance of ondansetron was increased and ondansetron blood concentrations were decreased.

Tramadol:

Data from small studies indicate that ondansetron may reduce the analgesic effect of tramadol.

Apomorphine:

Based on reports of profound hypotension and loss of consciousness when ondansetron was administered with apomorphine hydrochloride, concomitant use with apomorphine is contraindicated.

Incompatibility:

ONDATRON should not be administered in the same syringe or infusion as any other medication.

Pregnancy and Lactation:

Pregnancy

The use of ondansetron in pregnancy is not recommended.

In human epidemiological studies, an increase in orofacial clefts was observed in infants of women administered ondansetron during the first trimester of pregnancy. Regarding cardiac malformations, the epidemiological studies showed conflicting results.

Three epidemiological studies in the US assessed the risk of specific congenital anomalies, including orofacial clefts and cardiac malformations in offspring born to mothers exposed to ondansetron during the first trimester of pregnancy.

- One cohort study with 88,467 pregnancies exposed to ondansetron showed an increased risk of oral clefts (3 additional cases per 10,000 women treated, adjusted relative risk (RR), 1.24 (95% CI 1.03 - 1.48)) without an apparent increase in risk of cardiac malformations. A separately published subgroup analysis of 23,877 pregnancies exposed to intravenous ondansetron did not find an increased risk of either oral clefts or cardiac malformations.
- One case-control study using population-based birth defect registries with 23,200 cases across two datasets showed an increased risk of cleft palate in one dataset and no increased risk in the other dataset. There was no increased risk of cardiac malformations in this study.
- The second cohort study with 3,733 pregnancies exposed to ondansetron found an increased risk of ventricular septal defect, adjusted RR 1.7 (95% CI 1.0 - 2.9), but no statistically significant increase in risk of cardiac malformations.

Reproductive studies in rats and rabbits did not show evidence of harm to the fetus.

Pregnancy status should be verified for females of reproductive potential prior to starting the treatment with ONDATRON.

Females of reproductive potential should be advised that it is possible that ONDATRON can cause harm to the developing fetus. Sexually active females of reproductive potential are recommended to use effective contraception (methods that result in less than 1 % pregnancy rates) when using ONDATRON during the treatment and for two days after stopping treatment with ONDATRON.

Breastfeeding

Tests have shown that ondansetron passes into the milk of lactating animals. It is therefore recommended that mothers receiving ONDATRON should not breastfeed their babies

Adverse Effects / Undesirable Effects:

Immune system disorders

Rare : Immediate hypersensitivity reactions sometimes severe, including anaphylaxis.

Nervous system disorders

Very common: Headache.

Uncommon : Seizures, movement disorders including extrapyramidal reactions such as dystonic reactions, oculogyric crisis and dyskinesia have been observed without definitive evidence of persistent clinical sequelae.

Rare : Dizziness during I.V. administration, which in most cases is prevented or resolved by lengthening the infusion period.

Eye disorders

Rare : Transient visual disturbances (e.g. blurred vision) during I.V. administration.

Very rare : Transient blindness predominantly during intravenous administration. The majority of the blindness cases reported resolved within 20 minutes. Most patients had received chemotherapeutic agents, which included cisplatin. Some cases of transient blindness were reported as cortical in origin.

Cardiac disorders

Uncommon : Arrhythmias, chest pain with or without ST segment depression, bradycardia.

Vascular disorders

Common : Sensation of warmth or flushing.

Uncommon : Hypotension.

Respiratory, thoracic and mediastinal disorders

Uncommon : Hiccups.

Gastrointestinal disorders

Common : Constipation.

Hepatobiliary disorders

Uncommon : Asymptomatic increases in liver function tests (These events were observed commonly in patients receiving chemotherapy with cisplatin).

General disorders and administration site conditions

Common : Local I.V. injection site reactions.

Overdose and Treatment:

Little is known at present about overdosage with ondansetron. Reported symptoms include visual disturbances, severe constipation, hypotension and a vasovagal episode with transient second degree AV block. In all instances, the events resolved completely.

There is no specific antidote for ondansetron, therefore in all cases of suspected overdose, symptomatic and supportive therapy should be given as appropriate.

Compatibility with I.V. fluids:

In keeping with good pharmaceutical practice I.V. solutions should be prepared at the time of infusion. The diluted solution is a clear and colourless solution. ONDATRON should only be mixed with those infusion solutions which are recommended:

- Sodium Chloride Intravenous Infusion BP 0.9%w/v
- Glucose Intravenous Infusion BP 5%w/v
- Mannitol Intravenous Infusion BP 10%w/v
- Ringers Intravenous Infusion
- Potassium Chloride 0.15%w/v and Sodium Chloride 0.9%w/v Intravenous Infusion BP
- Potassium Chloride 0.15%w/v and Glucose 5%w/v Intravenous Infusion BP

Storage Condition:

Do not store above 30°C. Protect from light.

Shelf life:

3 years from manufacturing date in the proposed storage condition. Do not use after expiry.

Dosage form and packaging available:

2mL X 12 LDPE plastic ampoules per box

4mL X 20 LDPE plastic ampoules per box

Manufacturer / Product Registration Holder :

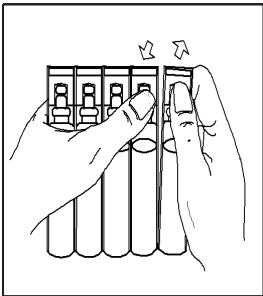


AIN MEDICARE SDN.BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

HANDLING INSTRUCTION

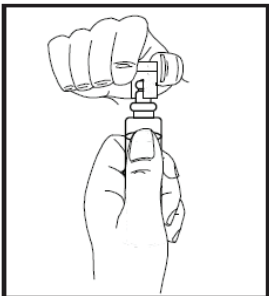
AIN MEDICARE INJECTION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) AMPOULE

1)



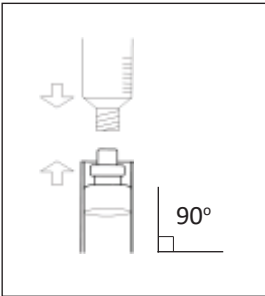
Detach ampoules along parting line

2)



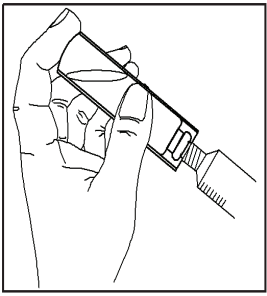
Twist off tab

3)



Attach luer lock syringe, luer slip syringe or hypodermic needle vertically to the ampoule.

4)



Draw out injection fluid using luer lock syringe, luer slip syringe or hypodermic needle. Ensure the connection is tightly secured for effective aspiration of the liquid.